SoI INNOVATION PROJECTS | AY2026-27

PHASE 1
Emergency Traffic Corridor

Provethe right customerand a significant pain pointbefore committing to a solution.

1 DISCOVER — 2 VALIDATE — 3 DECIDE

Rule: no fixed app, AI model, device or platform pitch in Phase 1.

Project snapshot

AIEmergencyTraffic Corridor

Emergencyvehicleslosecriticalminutesincongestedtrafficbecausesignals along the route are not clearedintime.

TEAM ID / TEAM NAME

NEURONIX |

BATCH / VERTICAL

2025–29 | AI&Data (AD)

MENTOR

CHITRA

PATHWAY

☐ Newproject
☐ Continue AY2025–26 project

CUSTOMER GAP & MATERIAL ADVANCEMENT

Gap:Emergency vehiclesget stuckinroute congestion and at red signals. Advancement: AI finds route congestion and coordinates signals into a green corridor, with human override.

STUDENTTEAMSUBMISSION - 10SLIDES

10-slide student submission | Due 30 September 2026 | Official review is recorded in the Assessment Workbook

2 Customer ecosystem and access

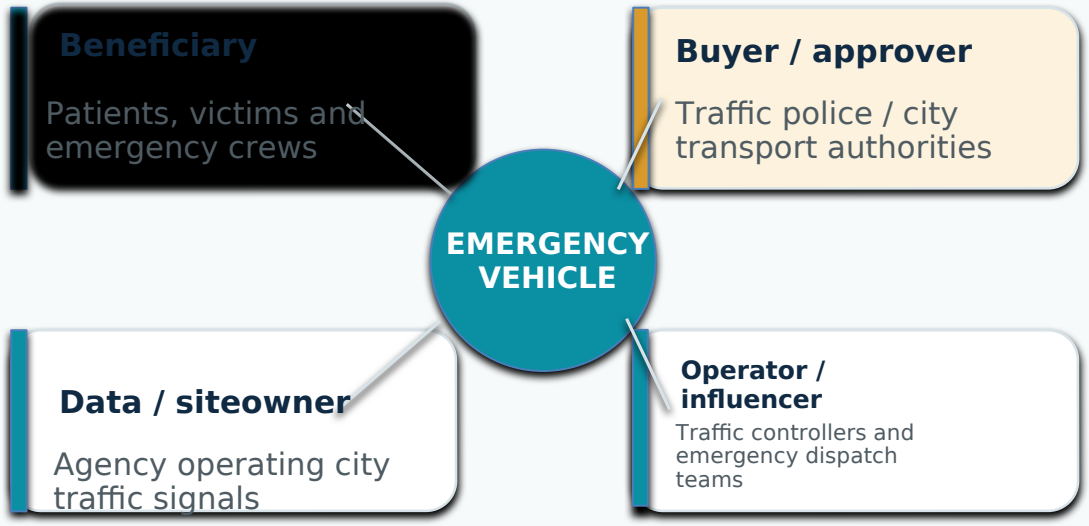
DISCOVER | VALIDATE | DECIDE

Identifywhoexperiencestheproblem,whobenefits,whodecides,andwhetherthe team can keep engaging them.

PRIMARY END USER

Ambulance / fire-engine drivers
City roads and signal junctions
During emergency response trips
Unpredictable congestion; time-critical

STAKEHOLDER MAP



ACCESS AND PERMISSION

Authorized traffic-police contact
Scheduled control-room visits
Signal access and permission required

REAL CUSTOMER SIGNAL

“[Insertverifiedquotefrom an emergency driverordispatcher]”

SEGMENT CHOICE

Emergency driversand dispatchers feel route delays directly.

The job they are trying to complete

Reach the incident or hospital quickly and safely by getting a clear path through traffic and signals.

3 Current workflow and pain moment

DISCOVER | VALIDATE | DECIDE

Show what happens today using a real recent occurrence—not a hypothetical future scenario.

1 TRIGGER

Emergency call is dispatched and the vehicle starts its route.

2 CURRENT ACTION

Driver uses siren and lanes, weaving through congestion and waiting at red signals.

3 WORKAROUND

Traffic police or dispatch manually clear junctions, or the driver reroutes.

4 OUTCOME / PAIN

Lostminutes; junction accident risk; delayed patient or incident response.

LAST REAL OCCURRENCE

Emergency route | Driver
Junction delay slowed response.

WHAT WORKS TODAY

Sirens, priority rules and manual traffic policing clear some routes.

WHAT REMAINS UNSOLVED

Congestion ahead is unknown and signals are not coordinated in advance.

VALIDATEDPROBLEM STATEMENT

Emergency vehicle drivers lose critical response time because congestion on the route is not known early and traffic signals are not cleared in time, causing delay

4 Customer validation plan and coverage

DISCOVER | VALIDATE | DECIDE

Planpurposefulevidencecollection.Quality,diversityandrelevancemattermorethancollectingmany convenient responses.

Stakeholder group	Planned / completed	Method	Why this group matters	Evidence IDs
Emergency vehicle drivers	Planned: 4+	Interview + route review	Directly face congestion and red signals	
Dispatchers / traffic controllers	Planned: 2–4	Interview + walkthrough	Explain dispatch, signal control and access	
Traffic police / city authority	Planned: 1–2	Stakeholder discussion	Tests approval, safety and adoption	

MINIMUM EVIDENCE FLOOR

NEW PROJECT
6+ meaningful conversations, including
4+ emergency vehicle drivers

METHOD MIX

- ☐ Probleminterviews
- ☐ Emergency route review
- ☐ Signal system study
- ☐ Secondary sources

SAMPLE AND ETHICS CHECK

- ☐ Different viewpoints
- ☐ Permission obtained
- ☐ Disagreement recorded
- ☐ Sensitive data anonymized

5 Evidence collected and key insights

DISCOVER | VALIDATE | DECIDE

Summarize the strongest evidence. Keep the original, anonymized proof in the linked evidence folder.

CONVERSATIONS
6+
4+ drivers
2+ otherstakeholders

OBSERVATIONS
6+
Congestedjunctions; red-signalwaits;manual clearing

BASELINE
TO VALIDATE
Emergency responsetimeon congested routes

SOURCES
4 TYPES
Interviews, routeobservations, traffic andsignal information, secondary sources

REPEATED INSIGHT

Unclear traffic aheadand signal delays slow response; the route must be cleared early.

CONTRADICTORY OR LIMITING EVIDENCE

Traffic varies by time andplace.Drivers may not yield; signal controlneeds authorization and oversight.

STRONGEST QUANTITATIVE SIGNAL

Noquantitativedata collected yet
Type:☐Measured ☐ Reported ☐ Estimated

What the evidence now says

Emergency vehicles lose time in congestion and at signals, and manual clearing is limited. Real-user validation and authorized traffic data are still required.

6 Pain significance, value and adoption

Showwhytheproblemdeserveseffortandwhetherarelevantdecision-makerhasareasontosupport improvement.

DISCOVER | VALIDATE | DECIDE

FREQUENCY
Every trip
Reported / to validate

SEVERITY
Potentially high
Patient and public safety

CONSEQUENCE
Time lost
Delayed care and rescue

VALUE
Responders
Faster route clearance

CURRENT ALTERNATIVES AND WHY USERS TOLERATE THEM
Sirens, priority-of-way rules and manual police clearing exist. The gappersists because congestion ahead and signal timing are not managed in advance.

ADOPTION / SUPPORT SIGNAL
Approvers: trafficpoliceand city authorities. Support signal: to be validated. Barriers: authorization, safety, reliability and human oversight.

BASELINE AND LATER SUCCESS TARGET
Baseline: siren and manual clearing
Improvement: faster,clearer route
Measure: trip time, junction delay, safety incidents

Phase 1 conclusion: The problem appears significant because congestion and signal delays can slow emergency response; actual impact must now be quantified with authorized traffic and response-time evidence.

7 Learning loop, problem boundary and remaining risks

DISCOVER | VALIDATE | DECIDE

A strong Phase 1 submission shows changed thinking and makes the remaining uncertainty explicit.

INITIAL BELIEF

Emergency vehicles lose time to congestion and signals that are hard to clear in advance.

EVIDENCE

Response relies on sirens and manual clearing. Constraint: traffic and signal variability.

REVISED FOCUS

PrioritizeAI congestion detection and signal coordination for a green corridor, with human override.

IN SCOPE

Congestion analysis; vehicle detection; signal coordination; human override.

OUT OF SCOPE FOR NOW

City-wideautonomous control, unsafe signal changes or unauthorized deployment.

KEY EVIDENCE GAP

Validatetime saved,safety, false triggers, signal reliability and authority acceptance using permitted data.

Assumption / risk	Current evidence	Next test / control
Customer / access / value	Weak	Interview drivers and dispatchers
Technical / data / resource	Medium	Test on permitted traffic data
Ethics / privacy / security	Weak	Review signal safety and access controls

Do not test on live roads, signals, vehicles or systems without permission and appropriate safety controls.

8 Vertical-specific Phase 1 checklist

DISCOVER | VALIDATE | DECIDE

Highlightyourverticalandaddressitscustomer,evidenceandresponsibilityquestions. Do not assume the preferred technology is necessary.

AD AI & Data

Human-in-the-loop signal control; manual baseline; authorized traffic data; safety, fairness and authority acceptance.

BW Blockchain & Web3

Actors/transactions; trust or reconciliation failure; governance/identity; why a database may be insufficient.

CD Cloud & DevOps

Developer/operator pain; deployment/incident baseline; reliability, cost, compliance and integration.

CS Cyber Security

Asset/user/threat/impact; current controls; legal authorization; safe test boundary.

EI Embedded & IoT

Physicalcontext; sensing/actuation need; power, connectivity, calibration, safety and maintenance.

FW Full Stack Web

End-to-end userjourney;data flow; roles/permissions; accessibility, privacy and adoption barriers.

MA Manufacturing & Automation

Operator/processobservation;cycle time/defect; ergonomics, safety, material, maintenance and manufacturability.

Team action: use evidence IDs for the highlighted AI & Data checklist; carry accuracy, false-trigger and data-access questions to Slide 9.

9 Our selected vertical: evidence and early success measures

DISCOVER | VALIDATE | DECIDE

Usethepromptsfrom Slide 8 to show that the problem is both customer-relevant and technically responsible within your lab.

SELECTED VERTICAL

AD
AI & DATA

Why this problem belongs in the chosen vertical

The problem needs real-time analysis of traffic data for time-critical signal decisions.

VERTICAL CUSTOMER / TECHNICAL CONTEXT

Congestion prediction; emergency vehicle detection; signal coordination; human override.

VERTICAL EVIDENCE ALREADY COLLECTED

Current workflow relies on sirens and manual clearing.

KEY CONSTRAINT / RESPONSIBILITY

Safety, authorization, reliability, false triggers and human oversight.

EARLY SUCCESS MEASURE

Baseline: siren and manual clearing
Target: shorter emergency trip time
Measure: trip time, accuracy, junction delay, safety

HIGHEST-RISK VERTICAL ASSUMPTION

AI may not predict congestion reliably enough to coordinate signals safely.
Next: authorized traffic-data tests.

PHASE 2 TECHNICAL LEARNING QUESTION

Can AI identify route congestion and coordinate signals into a green corridor reliably while preserving human oversight?

10 Phase 2 ideation preparation and next actions

DISCOVER | VALIDATE | DECIDE

Firstcompareexistingandnon-digitalalternatives;thenfreshlygenerateatleastthreematerially different ideas before selecting one.

FRESHIDEAS TOGENERATEINPHASE2 - NOT FINAL SOLUTIONS

A CONGESTION-AWARE ROUTING

Predict route congestion and suggest the fastest emergency path.

B SIGNAL COORDINATION

Coordinate signals ahead of the vehicle to create a green corridor.

C LIVE CORRIDOR ALERTS

Alert drivers and traffic control early about the cleared route and hazards.

CONCEPT REVIEWERS

Emergency drivers
Traffic authority
AI / Data specialist
[Number + date]

Concept selection criteria

Desirability | **Feasibility** | **Responsibility** | **Viability / value** | **User evidence**

Next action	Owner	Due date	Evidence / output
Study emergency route workflow	Team	[dd/mm]	Workflow analysis
Collect feedback + permitted traffic data	Team / AI member	[dd/mm]	Interviews + dataset
Test congestion and signal logic	AI member	[dd/mm]	Evaluation report

Phase 2 due 16 October 2026: compare current solutions, create 3+ fresh ideas, select one using Desirability, Feasibility, Viability/Value and Responsibility, and justify the trade-offs with evidence.

Official assessment and gate decision are completed by reviewers in the separate SoI Assessment Workbook; students do not self-approve.